TRACE-RPG: Source-Bound Ontology Evidence and a Bounded Codex-Assisted Integration Case in an Interactive Game

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***Abstract*—TRACE-RPG combines external authorization, typed-error repair, unchanged fallback, revalidation, replay, and accounting for generated game events. Its read-only bilingual ontology view binds source declarations to public observations without mutation authority. One operator-assisted Codex dashboard case (n=1, independent repeats 0) reports runtime `gpt-6.1-sol`, repaired geometry, retained failures, 62 checks, a Godot Web build, and real-input denial/disclosure/redisplay. Provider identity, revision, seed, cost, and inference latency remain unavailable. This bounded integration evidence is not model comparison, ontology superiority, or player benefit; traceability and whole-game/repeated-cross-model studies remain proposed.**

***Index Terms*—game AI, interactive narrative, symbolic validation, constrained generation, reproducibility**

I. Introduction

GENERATED events may require absent items, assert unknown facts, disclose secrets, introduce unauthorized effects, or regress quests. Plausibility [1], [2] and quest/entity grounding [3], [4] confer no authority: **only TRACE-RPG's external deterministic gate authorizes mutation** (Fig. 1).

C1—contracts, **C2—validate–repair–commit**, **C3—audit-linked evidence**, **C4—assignment accounting**, and **C5—guided repair** are bounded design/conformance surfaces (Table I). Active E1, E3, ENG1, NPC-DISC-1, and DEV-CASE evidence is not attributed to the development model. The new contribution is the source-bound read-only ontology view and single Codex integration package, not C6 or a full-game round trip.

II. Related Work and Research Gap

Grounding/believability spans quest/entity dialogue, graph quests, memory–reflection–planning, and trainable roles [3], [4], [5], [6]; player/dialogue and role studies separate admissibility from experience [7], [8], [9], [10]. Engagement research motivates future, unimplemented, non-authoritative affect [11], [12], [13].

State, embodied, agent/game-content, process, social, and long-horizon environments complement playtesting [1], [2], [14], [15], [16], [17], [18], [19], [20]; NCP-Bench measures narrative commitment/fact consistency, not player outcomes [21].

Structural constraints, critics, affordances, trigger rules, preconditions/effects, rule refinement, incremental validation, authored transformations, and JSONSchemaBench [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33] do not provide our transaction/audit/repair combination or superiority beyond encoded predicates. Retrieval/memory has no commit authority [34], [35], [36].

Reproducibility/evaluation motivate frozen artifacts, failure retention, construct/rater/bias controls, and justified margins/random effects [37], [38], [39], [40], [41], [42], [43], [44], [45]. We replay admitted transitions and use leakage checks for denominators, not efficacy [46], [47].

C1–C5 combine these established mechanisms [27], [28], [29], [48], [46], [49], [50], [51], [52], [53], [54]; no invention claim.

TABLE I
Contributions, Prior Topics, Evidence Lanes, and Inference Ceilings (Generated from the Structured Matrices)

ID	Contribution	Prior topics (refs)	Lanes	Inference ceiling
C1	Trust-boundary contracts	T1, T2, T3 (17)	E1, ENG1	encoded-field contract conformance
C2	Validate–repair–commit controller	T1, T8 (14)	E1, ENG1	mechanism behavior on frozen cases
C3	Audit-linked evidence layer	T4, T7, T8 (6)	E1, ENG1	integrity/replay on named fixtures
C4	Assignment-complete harness	T1, T4, T6, T7 (20)	E1	exact designed-case accounting
C5	Counterexample-guided operator ρ	T1, T8, T9 (8)	E1	frozen offline repairability classes

T1 = grounded game worlds; T2 = retrieval/memory agents; T3 = structured generation; T4 = environments/benchmarks; T5 = player experience; T6 = human/LLM evaluation; T7 = statistics/reproducibility; T8 = planning/runtime interposition; T9 = feedback repair. Lanes are defined in Table V. All rows are authored-fixture verified; none supports an efficacy claim.

III. Problem Definition and Guarantee Boundary

$c_t = (G_t, q_t)$ stores typed snapshot G_t and external action-policy/disclosure/quest-stage constraints q_t. Predicates neither read $h_{<t}$ nor reconstruct `WorldState` from logs; transitions update G_t, not q_t, and exclude policy rotation. Retrieval, scores, rationales, memory, and future affect have no authority.

Strict parsing of untrusted candidate a_t enforces the shared key/schema contract, rejecting unknown top-level fields. Validation returns

$$\begin{aligned} V(c_t, a_t) &= (\boldsymbol{v}, E_t), \\ \boldsymbol{v} &= (v_{\text{policy}}, v_{\text{pre}}, v_{\text{reach}}, v_{\text{know}}, \\ &\qquad v_{\text{disc}}, v_{\text{elig}}, v_{\text{stage}}) \in \{0,1\}^7, \end{aligned} \tag{1}$$

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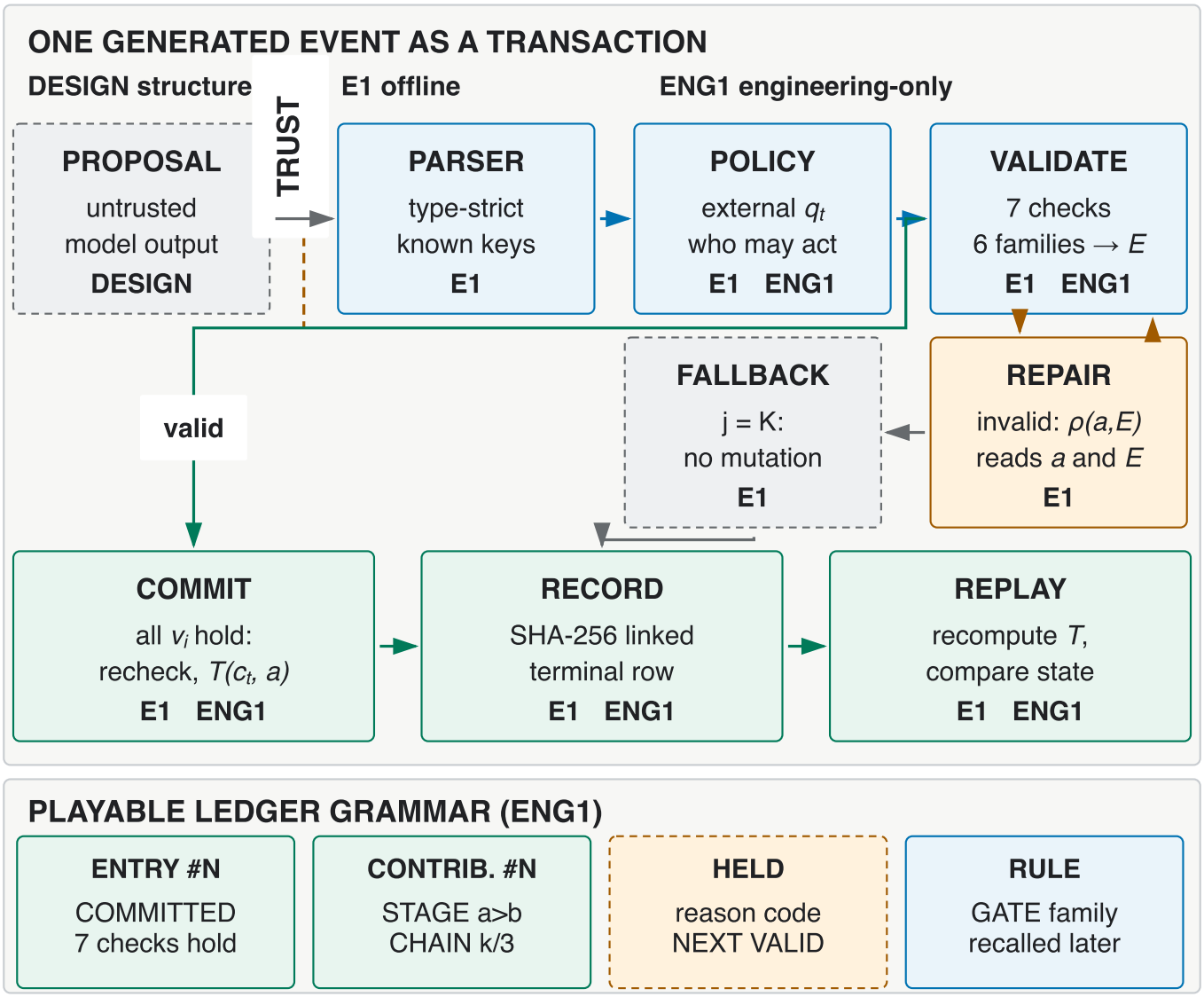


Fig. 1. Policy/c_t gate proposals; errors feed bounded ρ or unchanged fallback. Passing candidates are rechecked, committed, recorded, and replayed. Historical cohorts and legacy ENG1 remain archived, not active model evidence. Below: committed-snapshot ledger grammar.

where E_t is a structured error set. Mutation is

$$c_{t+1} = \begin{cases} T(c_t, a_t), & \bigwedge_{i=1}^{7}(v_i = 1), \\ c_t, & \text{otherwise.} \end{cases} \tag{2}$$

Fixtures test implementation invariants, not theorems. **I1:** validation exhaustion or pre-commit proposal/controller failure preserves state. **I2:** returning callbacks and $K \geq 0$ bound attempts to $K + 1$ with pre-commit revalidation; without enforced deadlines, wall-clock termination is not guaranteed. **I3:** accepted `CandidateAction.effects` stay within `ActionPolicy.allowed_effects`, including authorized quest-stage targets. **I4:** frozen schemas/deterministic transitions enable terminal-state reconstruction.

Assumptions: correct policy, faithful extraction, deterministic versions, returning callbacks, and single-process file ownership. Replay checks structured transitions, not repair provenance, prompts, stochastic generation, or undeclared natural-language implications.

IV. TRACE-RPG Architecture and Algorithms

A. Trust Boundary and Contracts

External models/adapters propose; parsing rejects ambiguous types/unknown keys; independent policy defines preconditions/effects. The versioned bridge carries observations, proposals, validations, rejections, fallbacks, and commits (Fig. 1); unknown-key parity establishes neither full parser equivalence nor commercial portability. Python has exactly seven deterministic checks with predicate IDs: `action_policy`, `precondition`, `object_reachability`, `npc_knowledge`, `forbidden_disclosure`, `quest_stage`, and `quest_stage_monotonicity`. Godot's six state-relative families and seven DISC rules are distinct, non-isomorphic

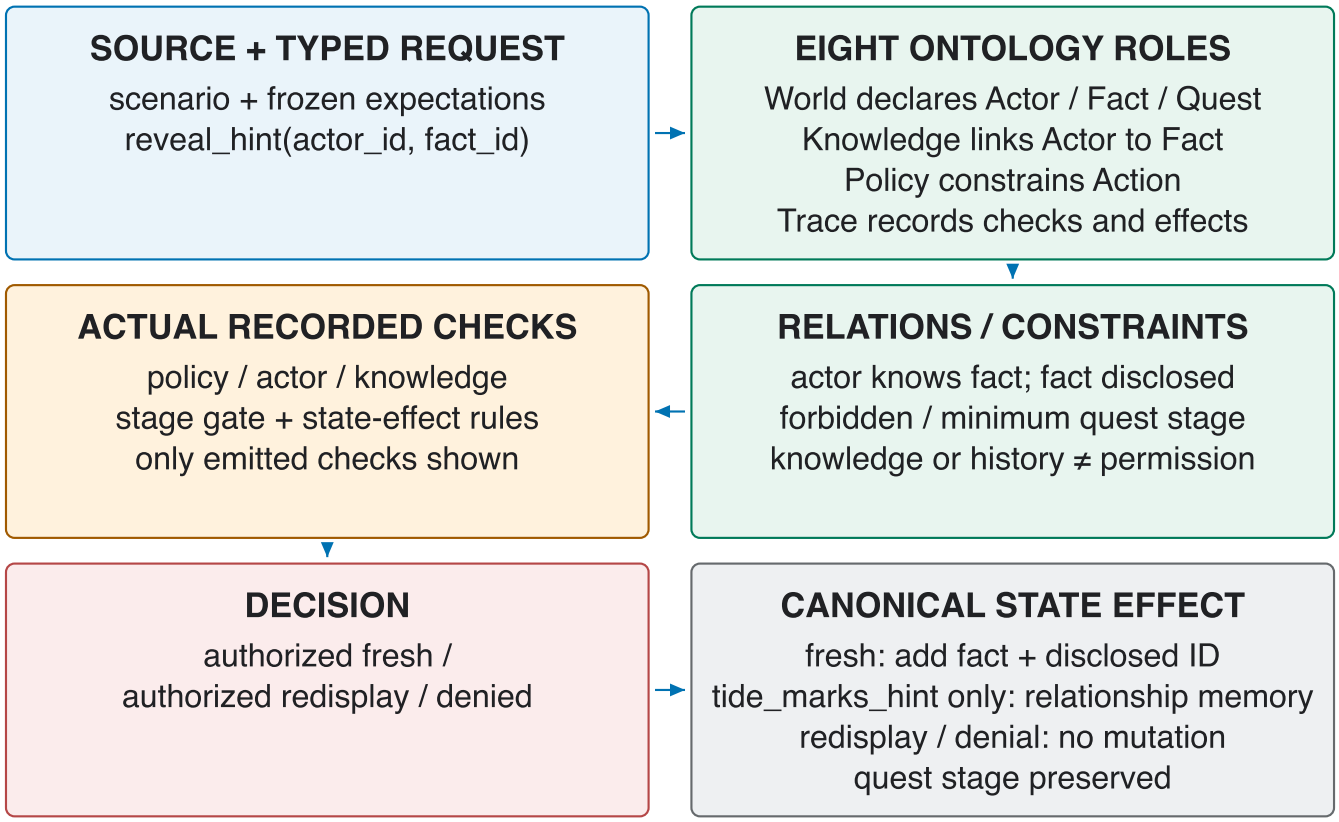


Fig. 2. Eight-role ontology structure (Table III) connects declared source fields to authored checks and observed decisions. Only authorized transitions change canonical state; the diagram is not live capture or hidden-state evidence.

TABLE II
ENCODED VALIDATOR BOUNDARY

Predicate	Rejects	On failure
Action policy	unknown action/type; omitted policy-required preconditions/effects; non-allowed effects or unauthorized quest-stage effects	unchanged
Precondition	absent/false requirement	unchanged
Reachability	inaccessible object	unchanged
NPC knowledge	undeclared known fact	unchanged
Disclosure	forbidden fact	unchanged
Quest stage (`quest_stage`)	required stage above current stage	unchanged
Quest-stage monotonicity (`quest_stage_monotonicity`)	regressive stage	unchanged

label sets; its single `QUEST STAGE` display family spans the two Python quest checks, while unauthorized quest-stage effects remain action-policy failures.

The role map `game-track/web/dashboard/ontology-map.json` binds declarations/frozen expectations; below, I abbreviates `initial_state`.

B. Fixed-goal NPC integration

Under `npc-goal-recovery/1.1.0`, Codex proposes intent, Python checks public claims/plans, and Godot alone writes state. The fixed harbor-restoration/authorized-tide-lead goal is not model-editable. Write bindings cover session, epoch, request, scenario, NPC, goal, version, policy/config hashes, revision/state hash, deadline/budget, and issued action. Service: loopback `127.0.0.1:8768`; browser origins: `localhost`, `127.0.0.1:8877`.

1) NPC-DISC-1 disclosure contract and authority boundary: NPC-DISC-1's 60 declaration-derived `reveal_hint` cases do not rerun/enlarge historical `verified-run-006`. Generated results/Tables X–XII define strata, bindings, and input-to-check mappings; Godot rechecks service bindings before mutation.

TABLE III
EIGHT ONTOLOGY ROLES: READ-ONLY SOURCE BINDINGS, NOT LIVE-STATE INFERENCE OR AUTHORITY

Role	Source field or relationship example
World	I.`world.object_locations`: `signal_lens`→`lamp_store`.
Actor	I.`npcs.captain_mira`; sole `scope.eligible_actor_ids` member.
Knowledge	Actor `knowledge`→canonical Fact IDs, not permission.
Fact	I.`facts`: `dock_saved`, `lighthouse_dark`; `keeper_betrayal` is an ID, not secret payload.
Policy	`disclosure_policy`: forbids `keeper_betrayal`; `tide_marks_hint` needs minimum quest stage 2.
Quest	I.`quest`: stage 0, flag `dock_saved`; disclosure preserves stage.
Action	`reveal_hint(actor_id, fact_id)` adds Fact/disclosed ID; only tide hint adds relationship memory. Fresh revision $+1$; redisplay unchanged.
Trace	`prior_state_hash`, `next_state_hash`, `rule_ids`, `template_id`, `case_id` bind packet/version to evidence; absent fields cannot establish PASS.

1: $a \leftarrow \text{Propose}(c)$; $j \leftarrow 0$
2: **repeat**
3: $(\text{ok}, E) \leftarrow V(c, a)$; $\text{Record}(a, E)$
4: **if** ok **then**
5: **assert** $V(c, a).\text{ok}$; **return** $\text{Commit}(c, a)$
6: **if** $j = K$ **then return** $\text{Fallback}(c)$
7: $a \leftarrow \text{Repair}(c, a, E)$; $j \leftarrow j + 1$
8: **until terminal**

Fig. 3. Validate–repair–commit procedure. At most $K + 1$ candidate attempts are recorded; success performs defensive validation before mutation.

TABLE IV
PER-ERROR-CLASS EDITS OF THE GUIDED OPERATOR ρ

Validator error class	Edit $\delta(e)$ on the candidate
Policy precondition omission	insert the named predicate
Policy effect omission	insert the named effect
Policy effect violation	drop the named effect
Unsatisfied precondition	drop the named predicate
Unauthorized stage mutation	clear `quest_stage_effect`
Unknown action type	no-op (no registered alternative)
Unreachable object; knowledge, disclosure, and stage classes	no-op (candidate-local edit would launder or needs state)

TABLE V
EVIDENCE LANES AT A GLANCE: DESIGN, UNIT, COMPARISON, AND CEILING (GENERATED FROM THE EXPERIMENT MATRIX)

Lane	Design	Unit / budget	Comparison	Ceiling
E1	Offline conformance: authored frozen fixtures, one world	13 gate; 12 repair/arm; 10 faults; 7 adapter; 3 guards	reject vs blind vs ρ vs oracle	mechanism conformance
E3	KG/ontology simulation: closed-world typed-link holdout	43 nodes; 106 ref. edges; 24 typed; 210 scores	degree baseline vs 6 fixed strategies	construction result only
ENG1	Engineering-only Godot/Web: engine-local policy mirror, no participants	4 fixtures; 52 checks; 8 smoke; 5 archetypes	authored fixture paths	engineering-only conformance

Active lanes are non-interchangeable with different denominators; the historical E2 receipts are excluded from active generation and ENG1 is engineering-only. No active lane establishes population efficacy, model ranking, player experience, retrieval benefit, or commercial-engine performance.

C. Validate–Repair–Commit

In Fig. 3's dependency-free pseudocode, Record appends in memory; persistence follows terminal consistency. (ok, E) abbreviates V with $\text{ok} := \bigwedge_{i=1}^{7}(v_i = 1)$; $V(c, a).\text{ok}$ revalidates.

Repair cannot mutate state; exhaustion preserves it. Typed `counterexample_guided_repair` reads only candidate a/errors E (its sole state-derived input), never `WorldState`:

$$\rho(a, E) = a \oplus \bigcup_{e \in E} \delta(e), \tag{3}$$

Table IV's $\oplus$ applies deterministic $\delta(e)$: one field/error entity, at most $|E|$/attempt; conflicting requirements/rejections leave fields unchanged. Fixed points require identical E, not convergence. The nondeployable state-reading oracle ignores errors, reconstructs preconditions/effects/both quest-stage fields from state, and never launders violations by deleting required objects/used facts/disclosed facts.

D. Integrity, Semantic Replay, and Accounting

Unkeyed SHA-256 checksums/trace links detect accidental/tested changes, not writer identity or joint rewriting. Replay parses prior state/candidate/validation, requires invalid nonfinal attempts, recomputes final state, checks equality/JSONL continuity, and never reruns repair. Request→decision→result and first-issued-action/exactly-one-receipt binding do not prevent duplicate IDs or impersonation. Terminal classes are commit, symbolic fallback, adapter failure, and controller failure; the latter preserves state/assignment without a completed proposal trace. Pre-execution keys bind run, arm, scenario, seed, model, revision, configuration/input/prior-state hashes; aggregation rejects duplicate/missing/unexpected keys and separates failures.

V. EVIDENCE LANES AND BOUNDARIES

NPC-DISC-1 disclosure counts and DEV-CASE dashboard observations are additional typed lanes, not pooled with E1/E3/ENG1.

A. *E1* Offline Conformance Design

PQ1–PQ5 cover immutability, frozen-class repair, checksum/semantic/linkage/continuity faults, second-append rollback, fail-closed assignment/adapter/manifest accounting, and proposal/replay unknown-key rejection. Generated results enumerate 13 gate fixtures (1 valid, 12 negative) and 12 repair fixtures with classes; latency/token constants test accounting, not population or detector performance. Packets bind fixtures, assignments, schemas, results, and code.

B. Archived live-screening materials

Archived live-screening is immutable provenance/hash evidence, excluded from active tables/counts/claims and never transferred to `gpt-6.1-sol`.

TABLE VI
SIMULATION-ONLY PROPOSAL-STRATEGY EVALUATION ON A FROZEN TYPED-LINK HOLDOUT

Strategy	Precision	Recall	MRR	Brier	Sem@3
S0 degree baseline	0.000	0.000	0.282	0.315	0.556
S2 typed-lexical	1.000	1.000	0.944	0.131	1.000

TABLE VII
E1 OFFLINE CONFORMANCE: EXACT REPAIR-ARM COUNTS BY REPAIRABILITY CLASS (DESIGNED FIXTURES)

Arm	**Info source**	**Commits / cases**			**Mean edits**[e]
		G-rep.[f]	**O-only**[f]	**Irrep.**[f]	**Fields**
Rejection-only (K=0)	none	0/5	0/1	0/6	—
Blind unchanged retry	none	0/5	0/1	0/6	—
Validator-error-guided ρ	error set E	5/5	0/1	0/6	1.0
State-reading oracle	state+policy	5/5	1/1	0/6	1.0

[e] Mean changed candidate fields over committed repairs (minimality proxy); failed arm executions left the state unchanged. E1 uses 12 initially invalid designed cases: rejection-only/blind 0/12, 0/12; validator-error-guided ρ 5/12; state-reading oracle 6/12 commits. [f] Commits/cases. G-rep. = guided-repairable, O-only = oracle-only, Irrep. = irreparable (frozen repairability classes); these are not population efficacy or live-model quality.

C. NPC-ENG1/NPC-SCR1: 88 Assignments, 28 Definitions

Stable v5 has 16 engineering assignments and 12 screening recipes ($12 \times 2 \times 3 = 72$; combined $R/P/O = 40/24/24$). Checks cover bootstrap, identity/hash, intent/plan/role/result, negative/stale/duplicate/deadline, and replay. P uses authenticated Codex OAuth without fallback; O is 100% auxiliary-source, not independent gold. Status/count/latency/reproduction: `research/academic-pipeline/npc-goal-recovery/verified-run-006/`; verified-run-002/003/004 are historical. ENG-X01/SCR-X01, D02/QA-001, C-RESULT, Frozen8/Frozen16, v1–v4, and `native012` are provenance, not C6 evidence.

D. *E3* KG/Ontology Link-Proposal Simulation

E3 is an engineering-only, non-authoritative authored closed-world simulation. Six graph-edge-withheld authored targets select/summarize among seven fixed strategies, not an untouched or external test. **Scope.** This table reports authored closed-world link recovery, not runtime KG retrieval, semantic completeness, long-horizon contradiction reduction, or evidence for any efficacy claim. The deterministic methods graph contains 43 graph nodes, 106 reference edges, and 24 curated typed edges; all ontology checks pass. The generated table is construction evidence only; it establishes neither OWL/SHACL conformance, runtime retrieval, semantic completeness, nor efficacy.

TABLE VIII
E1 OFFLINE CONFORMANCE: FROZEN-PILOT CHECKS AND TERMINAL OUTCOMES (ROWS ARE NOT COMPARABLE)

Check or terminal family	**Exact outcome**	**Interpretation**
Gate fixture agreement	13/13	authored-oracle agreement
Known-boundary acceptance	2/2	encoded limit, not a safety pass
Closed unknown-key rejection	1/1	proposal/replay parser parity
Detectable integrity-fault rejection	10/10	designated check operations
Known provenance-boundary acceptance	1/1	repair provenance unauthenticated
Adapter terminal classes	1 commit; 1 fallback; 5 failures / 7 assigned	complete classification of 7 assignments
Manifest-guard rejection	3/3	injected faults failed closed

Rows have different denominators and favorable directions; they must not be ranked as rates. Every value is an exact audit fact for designed fixtures.

VI. PILOT RESULTS

A. Declared-Field Gate Conformance

The pilot loader admits a fixture set only when it isolates every implemented validator code exactly once alongside one valid control, so the 13/13 agreement and the observation of all 12 implemented error codes are *construction invariants of the harness*, not measured success rates. What the run adds is the part the loader does not enforce: each of the 12 isolated negative fixtures showed observed agreement with its authored expected code rather than reaching a different one, and every noncommit path preserved the complete canonical state. This is implementation conformance to an authored structured-field oracle over a single world state, not accuracy against independent semantic labels.

B. Repair-Arm Comparison on Designed Fixtures

The 12 initially invalid designed cases are partitioned into frozen repairability classes: 5 guided-repairable, 1 oracle-only, and 6 irreparable. Rejection-only and blind unchanged-candidate retry each committed 0/12. The validator-error-guided operator ρ committed 5/5 guided-repairable cases and, by design, 0/1 oracle-only and 0/6 irreparable cases; every ρ commit edited a mean of 1.0 candidate fields. ρ reads only the prior candidate and typed validator errors E; those errors are its sole state-derived input, and every failed arm execution left the canonical state unchanged. The state-reading reference callback committed 5/5 guided-repairable and 1/1 oracle-only cases, with 0/6 irreparable. These exact counts separate the three mechanisms on the designed fixtures: blind retry never converts a validator error into a commit, ρ converts exactly the cases whose error payload carries sufficient information, and the oracle additionally converts the case requiring state knowledge. The 5 committed guided cases initially carried `POLICY_PRECONDITION_OMISSION+UNSATISFIED_PRECONDITION`, `POLICY_EFFECT_OMISSION`, `POLICY_EFFECT_VIOLATION`, `UNSATISFIED_PRECONDITION`, `POLICY_QUEST_STAGE_EFFECT_VIOLATION`, respectively, and each

became valid through the corresponding Table IV edit $\delta(e)$ alone. This is operator conformance on frozen fixtures, not repair quality of any live model.

C. Integrity, Boundaries, and Assignment Accounting

All 10 faults prespecified as detectable were rejected by their designated check operations (10/10). Because this pilot does not compare stable typed detector codes, it does not establish detector-layer attribution. Separately, the known provenance-boundary fixture substituted a different invalid precursor before the same recorded valid repair, recomputed the checksum, and passed replay as expected (1/1 undetected). Thus, the mechanism does not protect against a writer who can recompute unkeyed checksums, and replay does not authenticate the repair-generation operation. Two open boundary sentinels—unextracted narrative disclosure and simultaneous omission of a required object from candidate and policy—were intentionally accepted at the encoded layer (2/2), with 0 labelled as safety passes. The former unknown-top-level-field acceptance boundary was later reclassified as a closed negative regression: both proposal and replay parsers specifically rejected a complete 12-field candidate carrying one unknown key (1/1). This establishes parser rejection parity, not semantic safety. Among seven adapter/accounting assignments, outcomes were 1 commit, 1 symbolic fallback, and 5 adapter failures. Synthetic telemetry fields were populated and propagated for 2/7 assignments; these are schema-pinned constants verifying accounting-field propagation, not measurements. All 3/3 injected duplicate or missing-assignment guards failed closed. These are raw counts from synthetic offline fixtures.

D. Fixed-Goal NPC Operation (NPC-ENG1/NPC-SCR1)

Mira's fixed goal is to deliver the authorized tide lead after harbor-signal restoration. Python computes bounded BFS plans from typed intent and public observations. Acquire/install remain player actions; the NPC may only commit the authorized `reveal_hint`. Godot synchronously checks session/epoch/request/goal/policy/revision/hash before calling the existing writer without an await. Results and new observations determine the next plan, unlike single-candidate validator-error repair.

NPC-ENG1 comprises 16 deterministic R-arm engineering cases initialized from actual engine snapshots (assigned_failure=2, completed=14). NPC-SCR1 retains all 72 assignments and failures, 24 per P/R/O arm (assigned_failure=12, completed=60). P records 24 authenticated Codex CLI exec attempts, 24 returned decisions, and 4 assigned failures; these are not remote billing-call counts. Requested model: unknown (null); recorded decision-model identifier: unknown (null). The latter propagates the configured selector, not a provider revision; the provider revision is unavailable. Null identifiers remain unknown; global defaults are not evidence. R uses authored rules. O is a protocol-declared auxiliary under `same_model_family_auxiliary`. Actual model-family matching is unverified. No source is independent; O is neither an oracle nor semantic gold. Table IX reports

TABLE IX
NPC-SCR1 FIXED-CASE EXECUTION COUNTS, SEPARATE FROM THE 16 NPC-ENG1 CASES; ALL ASSIGNED FAILURES REMAIN IN THE DENOMINATOR.

Arm	Assigned	Complete/fail	Goal reached	Recovery E/U
P	24	20/4	4	24/0
R	24	20/4	4	24/0
O	24	20/4	4	24/0

E: recovery eligible; U: unreachable; NA: unmeasured. Observed goal counts are not recovery rates. Valid replay does not turn a failed assignment into a completed one. O has zero independent sources.

bounded protocol-conformance fixed-case counts, not recovery-efficacy, superiority, or generalization estimates. Efficacy remains pending; tokens and cost are unmeasured. A read-only CLI and isolated working directory were used, but tool-free inference and adversarial confidentiality were not established. This generation revalidates immutable verified-run-006 against archived source bytes dated 2026-10-17 (revision fa9fb8a). It is a historical packet result and does not certify the current checkout NPC runtime or current release authority. Verified pre-authored stale-control ACK rejections are P=4, R=4, O=4 per arm (SCR-T01 (stale_revision), SCR-T02 (stale_policy)); 12 instances had all corresponding negative-control gates passed. These rows remain assigned failures in the denominator, not method-performance outcomes; other assigned failures (0) remain unclassified. Replay validation is limited to the exported epoch (`intent_lineage_scope=epoch_local_only`); it does not prove intent continuity across epochs (`cross_epoch_intent_continuity_proven=false`). The opaque-anchor exception is restricted to an adjacent leading observation request/decision pair in a positive-epoch snapshot. Epoch-zero orphans, other observation requests before intent commitment, and later intent mismatches remain rejected.

E. Frozen `reveal_hint` contract and disclosure evidence

This artifact consumes only the frozen expectation packet `sl-npc-disclosure-expectations-v1` (v1.0.0) and four actual result files. The packet SHA-256 is `9920aa834910a3ee35ed4446de4fdba42bb67dda71b07a93bccfc7a0b2e09028`, checked directly against its sidecar. The declared normal stratum has 48 cases from `1 eligible actor x 4 declared fact IDs x 3 declared stage values x 2 knowledge values x 2 prior-disclosed values`; boundary has 4 cases and malformed inputs have 8 cases. Boundary and malformed inputs are excluded from the normal denominator, and each trace expectation is matched back to the packet by case ID rather than copied from a gate output.

The content-first explanation reads typed input (operation, actor, fact, stage, knowledge, and prior disclosure) into ontology entities/relations, applies authored rules, records actual checks, then reports an allowed/denied decision and its canonical state effect. A permitted first display adds only the requested

fact (plus the declared relationship-memory entry for the tide hint), preserves stage, and increments revision once. Denials and redisplays preserve canonical state. The presentation is explanatory: it does not recover hidden facts or natural-language implications and does not claim live capture.

The runtime declaration is `sl-npc-disclosure-expectations-v1:runtime-disclosure` / v1.0.0, with provenance `frozen_expectation_packet`. Scope contains four canonical fact IDs (dock_saved, keeper_betrayal, lighthouse_dark, tide_marks_hint) and the eligible actor `captain_mira`. Rule IDs are DISC-001, DISC-002, DISC-003, DISC-004, DISC-005, DISC-006, DISC-007; the only approved templates are `mira-safe-deflection-v1` and `sl-rc-003-authorized-hint`.

The Python evaluator materializes the same cases and expectation object in three isolated arms and records semantic outcomes, check traces, errors, before/after state SHA-256 values, rendered template IDs, and replay equality. Godot is the separate game-execution boundary that owns the existing `reveal_hint` writer and canonical mutation; this producer neither reconstructs nor replaces Godot state. The mapping is traceable through operation `reveal_hint`, scenario `sealed-lighthouse-v1`, actor `captain_mira`, and bound source hashes. Execution-observed hashes match the current source. 3 source hash(es) differ from the frozen baseline (game-track/godot/scripts/game3d/game_3d.gd, game-track/godot/scripts/sealed_lighthouse_machine.gd, llm-wiki/wiki/reports/2026-10-22-ontology-signals-interview.md); the packet hash is preserved and the difference is disclosed.

Table XI reports the three arms over the same 60 cases with separate normal, boundary, and malformed denominators. Violations in unchecked are intentional isolated observations. The checked general-rule and ontology-gate arms have identical outcomes on all 60 of 60 authored cases. The rule-change fixture set contains no discriminating case between these arms: all 2 fixtures and their 4 paired cases match. This is an authorization-conformance limit under the shared frozen rules, not evidence of general ontology equivalence or ontology superiority. Figure 4 distinguishes expected-match/replay case counts (E/R), check-record PASS/FAIL counts (C), and counts of cases with no reported checks (NR). NR does not mean skipped, pass, or out-of-scope. Table XII maps the rule-change fixtures and paired cases to disposable policy clones; the shipped scenario is not modified.

Claims are limited to bounded engineering conformance. This evidence does not establish human gold labels, an access-separated holdout, efficacy, superiority, generalization, player benefit, model latent state, or cost. Exact input paths, hashes, and reproduction commands are bound by `paper/latex/generated/disclosure_producer_manifest.json`. Disclosure `t_ms`/hashes/checks are engine-timeline signals, not provider receipt time, inference latency, or efficacy; missing differs from out-of-scope.

ENG1. Legacy Godot 4.7.1 replays authored rejection/commit to prior/offline hashes; presentation captures show board/hints/coach/chain prompt and dashboard checks. Core 52/52 plus four extensions establish engine-local conformance, not transport, participants, narrative, usability, or efficacy.

TABLE X
FROZEN REVEAL_HINT CONTRACT STRUCTURE, CONTENT, RULE IDS, AND RESULT PROVENANCE.

Field	Value
Operation	reveal_hint
Scenario	sealed-lighthouse-v1
Normal denominator	48
Boundary	4
Malformed	8
Canonical facts	dock_saved, keeper_betrayal, lighthouse_dark, tide_marks_hint
Eligible actor	captain_mira
Ontology	sl-npc-disclosure-expectations-v1:runtime-disclosure
Rule IDs	DISC-001, DISC-002, DISC-003, DISC-004, DISC-005, DISC-006, DISC-007
Templates	mira-safe-deflection-v1; sl-rc-003-authorized-hint

Expectations are not derived from gate output; denominators are read from the frozen declaration.

F. Single-task Codex development observation

Frozen `ontology-map-20261023-01` covers one read-only bilingual dashboard task (n=1, independent repeats 0), not an NPC trial/cohort expansion. Six native CLI invocations (initial + five resumes; one read-only/no-edit) are one case. `turn_context.model=gpt-6.1-sol` reports runtime selection, not provider attestation; provider identity/revision/seed/cost are `null`, model-only latency unavailable. Wall-clock 5942.665 s and CLI 263.32 s are neither inference latency nor the predeclared future endpoint.

Invocation six fixed inherited 660 px mobile-table minimum width: independent geometry RED (2 failed/35 passed), then 62/62 map/signal/Web checks including cell/paragraph geometry and a successful Godot Web build. Real-input browser attempt 10 passed; 01–08 failed setup/routes, 09 passed older gates with clipping. Final $(\text{seq}, t_{\text{ms}})$: stage-0 denial $(68, 19468)$ (equal hashes), fresh disclosure $(239, 58234)$ (different), authorized redisplay $(249, 59793)$ (equal). Observed 1440×900/390×900, keyboard Enter, inspection/frozen-comparison invariance, and zero pre-navigation console/page/network errors or warnings. Dated QA10 PASS differs from additive overall **NOT_PASS**: seven component gates passed; pointer-lock/yaw-guard and four pointer-lock page errors remained. This bounded integration is neither full-game creation, Python–Godot authorization round trip, nor model outcome. Privacy/source-binding failures and same-thread repairs add no n; Jev is `proposal_only`, without label/approval authority. Report/contract: `research/development-cases/ontology-map-20261023-01/result.json`, `contract.json`.

VII. DISCUSSION, THREATS, ETHICS, AND ARTIFACT SCOPE

Semantic repair [48], [55], [56], [57] addresses iterative correction; ρ uses deterministic pre-execution errors and oracle-only handles state reading. Frozen-case rule agreement and edit-scope diagnostics are neither capability nor ontology superiority.

TABLE XI
RAW-TRACE OBSERVATIONS FOR THE THREE DISCLOSURE ARMS WITH SEPARATE NORMAL, BOUNDARY, AND MALFORMED DENOMINATORS.

Arm	n (N/B/M)	Allow/Deny	Semantic mismatch	State/Display fail	Replay fail	Unique unsafe
unchecked_isolated	48/4/8	60/0	60	0/0	0	51
general_rule	48/4/8	16/44	0	0/0	0	0
ontology_gate	48/4/8	16/44	0	0/0	0	0

N/B/M = normal/boundary/malformed. Semantic mismatch compares the complete expected outcome; unique unsafe cases are the union of unauthorized allows and redisplay mutations. Permitted fresh-disclosure representation mismatches alone are not unsafe. Unchecked observations do not establish efficacy, superiority, or generalization.

TABLE XII
RULE-CHANGE FIXTURE TRACEABILITY AND EDIT SCOPE FOR PAIRED CASES; THE SHIPPED SCENARIO IS NOT MODIFIED.

Fixture	Mutation	Paired cases	Changed paths	Missed cases	Match
RC-001-stage-gate-2-to-1	disclosure_policy.stage_gates[0].minimum_stage 2→1	N-043, N-044	disclosure_policy.stage_gates[0].minimum_stage	none	PASS
RC-002-forbid-tide-marks-hint	disclosure_policy.permanently_forbidden ['keeper_betrayal']→['keeper_betrayal', 'tide_marks_hint']	N-047, N-048	disclosure_policy.permanently_forbidden[1]	none	PASS

Each fixture applies the same mutation to general_rule and ontology_gate. PASS requires no missed case and mutation_scope_preserved=true.

Unexecuted plans cover matched-information whole-game comparisons, repeated cross-model runs with exact revisions, independent semantic oracles, and a justified blinded human study (locators below). The future endpoint is time-to-first-all-predeclared-validation-pass, from matched request/evidence receipt to the first receipt passing every frozen mandatory check; unfinished/timeout/failure/stage-not-reached runs have no successful time.

A. Proposed traceability instrument (not measured)

The proposed construct is recoverable requirement/artifact/action/outcome correspondence, not correctness, hidden reasoning, developer skill, or player benefit. Freeze neutral U_d/links $L_d(u)$; independent assessors award 1 only after exact source/version, every raw-evidence relation, and a locator are verified. Report n_d/N_d, disagreement, and seeded-corruption controls, not weighted composites; incomplete inventories are indeterminate. Missing/inaccessible/stale/contradicted/unverified links earn 0 with reasons; verified zero repairs may be N/A, but empty logs are not zero. Report PASS, FAIL, NOT_RUN, STAGE_NOT_REACHED, OUT_OF_SCOPE, and MISSING separately; `stage_not_reached` is never PASS.

Threats and boundaries. No comparator, repeats, neutral complete-attempt inventory, or independent traceability scores exist. State consistency alone establishes neither behavioral nor narrative coherence [58]. Predicate/role/Godot/DISC sets are not latent-state vectors; declaration expectations are not human gold/holdouts, and historical/active lanes keep separate denominators. No participant/affect/comprehension/agency/enjoyment/immersion/benefit, population, safety-rate, model-ranking, or commercial-performance inference follows. Hashes do not authenticate; replay equality does not prove semantic correctness. Future human/affect studies require ethics review, consent, compensation disclosure, minimization, retention/deletion, opt-out, and calibrated judges [40], [41], [42], [43].

TABLE XIII
PROPOSED SOURCE-DERIVED TRACEABILITY DIMENSIONS (UNMEASURED)

ID	Required correspondence	Unit numerator / denominator
TR01	identity/activity binding to task, exact artifact revision/digest, activity, and actor	complete slots / required slots
TR02	request/input → rule version → evaluated check/result → decision	complete chains / inventoried decisions
TR03	decision → observed state or artifact effect with before/after evidence	complete correspondences / decisions requiring outcome
TR04	triggering failure → corrective package → exact revalidation → result	complete repair chains / actual repair attempts
TR05	forward requirement → artifact → verification and reverse in-scope change mapping	report forward and reverse separately
TR06	non-author independent exercise of declared audit/replay target	exercised targets / declared targets

Proposed and unmeasured; codebooks/source rationale: Data and Code Availability. No score was collected or validated.

VIII. CONCLUSION

TRACE-RPG combines bounded C1–C5 authorization/repair/replay/accounting with a source-bound bilingual observer. The single Codex case retains failures and acceptance boundaries, not model-performance evidence. Whole-game, timing, traceability, and repeated-cross-model evaluation remain future work; historical cohorts stay unpooled.

DISCLOSURE OF AI USAGE

LLMs assisted prose/code/tests/citations; hosted candidates were labelled. Curated AI art/desk mesh has file provenance; rendered snapshots/hashes/ledger are presentation-only. Authors checked sources and accept responsibility; NPC inputs/tables retain assignment/protocol/schema hashes and unknown token/cost/model-revision fields.

Data and Code Availability

Implementation/build source, frozen packets, Godot slice, deterministic evaluation, and machine-checked contribution, reference-topic, and experiment matrices: research/academic-pipeline/. Proposed rubric/codebook/source manifest: research/traceability-scoring-20261023/; future whole-game/repeated-cross-model plan: research/development-comparison-20261023/. Frozen disclosure inputs: paper/latex/generated/disclosure_producer_manifest.json. Standalone artifacts: paper/latex/figures/fig_signal_comparison.pdf, paper/latex/figures/fig_npc_screening.pdf; in-paper signal diagram/three-arm table have distinct roles. Earlier directories are historical receipts, not current-runtime proof; no reviewer archive/DOI deposit is claimed.

EXTERNAL TRACE COUNTS

	N normal	B boundary	M malformed
U	E 0/48 n=48 R 48/0 C 0/0 NR 48	E 0/4 n=4 R 4/0 C 0/0 NR 4	E 0/8 n=8 R 8/0 C 0/0 NR 8
G	E 48/0 n=48 R 48/0 C 14/68 NR 0	E 4/0 n=4 R 4/0 C 2/4 NR 0	E 8/0 n=8 R 8/0 C 0/16 NR 0
O	E 48/0 n=48 R 48/0 C 14/68 NR 0	E 4/0 n=4 R 4/0 C 2/4 NR 0	E 8/0 n=8 R 8/0 C 0/16 NR 0

U unchecked_isolated · G general_rule · O ontology_gate

E expected P/F (n cases) · R replay P/F

C records P/F · NR cases with no reported checks

P pass · F fail; NR unreported, not pass/skipped/out-of-scope.

Fig. 4. NPC-DISC-1 counts by arm/stratum. E/R: expected-match/replay; C: policy-check PASS/FAIL; NR: no reported check, not skipped. Packet-bound engineering counts do not measure latent state, free-language leakage, ontology superiority, or player benefit.

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